

WASP-12b

R-0394

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_230128 · created 2026-07-19T01:47:41+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459973.71183 +/- 0.00049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1426 +/- 0.0062
Transit depth (Rp/Rs)^2	2.03 +/- 0.18 [%]
Orbital Inclination (inc)	86.79 +/- 0.62
Airmass coefficient 1 (a1)	1.2139 +/- 0.0088
Airmass coefficient 2 (a2)	-0.177 +/- 0.02
Scatter in the residuals of the lightcurve fit is	0.73 %
Best Comparison Star	None
Optimal Aperture	4.51
Optimal Annulus	16.13
Transit Duration (day)	0.13252 +/- 0.00091

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1426 ± 0.0062 vs 0.1178 (deviation 2.88σ)
Duration fit vs published	0.13252 d vs 0.125 d (deviation 5.96σ)
Mid-time O–C	1568.4 min
Depth SNR	16.6
Residual scatter	0.73 %

Light curve

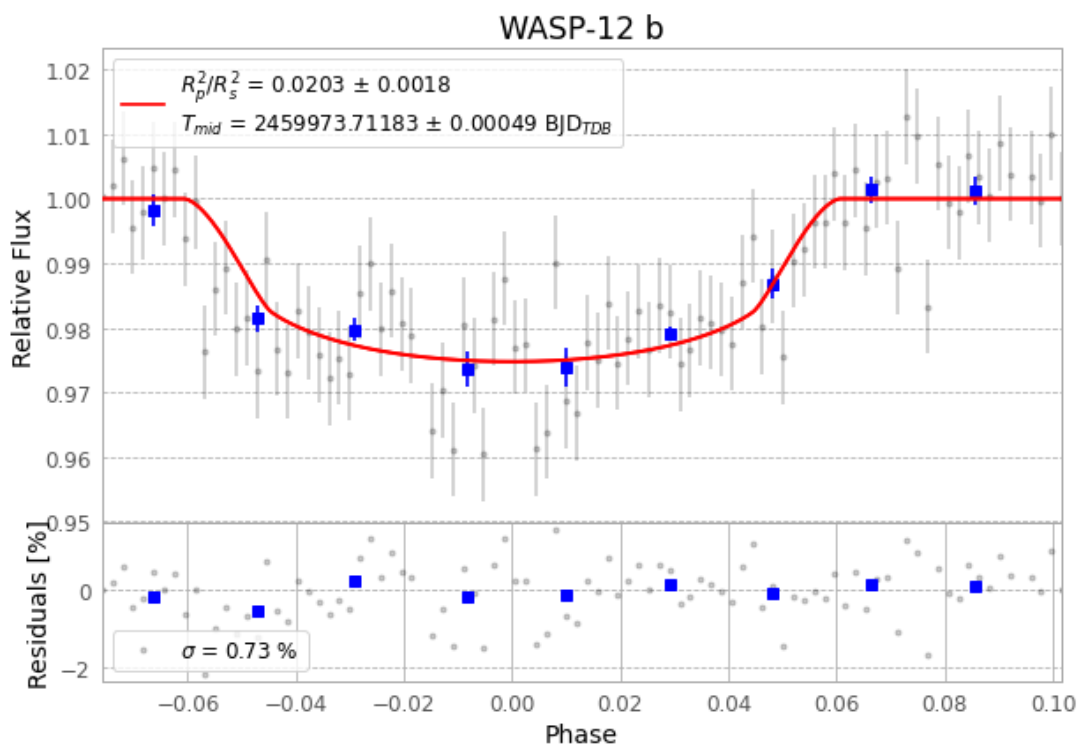


Figure 1 — FinalLightCurve_WASP-12 b_2023-01-28.png (SHA-256 53c76f64822473ac...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [433, 136], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 06a61cf9f73c910a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

94 FITS frames (62 MB), WASP-12230129025715.FITS → WASP-12230129073615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230128.log (1fefc33d3ed9...), FinalLightCurve_WASP-12 b_2023-01-28.png (53c76f648224...), FinalLightCurve_WASP-12 b_2023-01-28.csv (cc6b2c572191...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 01:47Z.